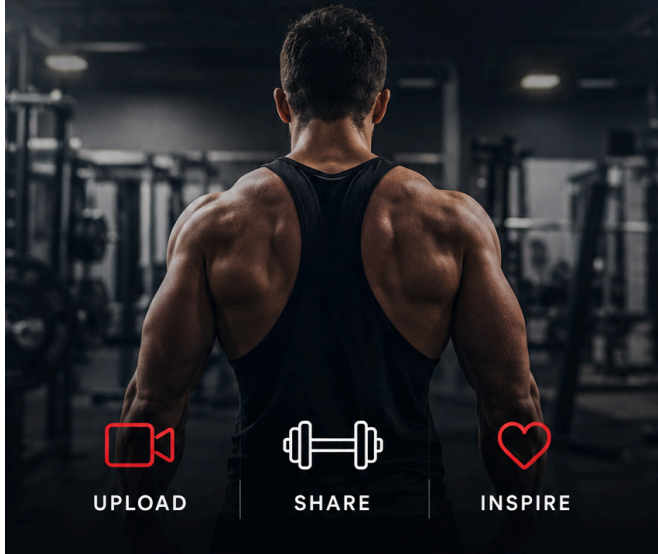


9:41



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Tech-Stack

Frontend: React Native + Expo — Cross-platform mobile application for iOS and Android.

Backend: FastAPI (Python) — REST APIs that handle authentication, business logic, and AI processing.

Database: Supabase (PostgreSQL) — Stores user accounts, videos, profiles, and application data.

AI: YOLO + Moderation Pipeline — Computer vision analyzes uploaded videos for object detection and content moderation.

Development Environment

- Cursor (AI IDE) – AI-assisted software development using Claude for code generation, refactoring, debugging, and architecture design.
- Git & GitHub – Source control, version management, and collaboration.

Frontend

- React Native – Component-based mobile application framework for building native iOS and Android applications.
- Expo – React Native platform that simplifies development, testing, and deployment with a single cross-platform codebase.
- TypeScript – Strongly typed JavaScript used to improve maintainability and reduce runtime errors.

Backend

- FastAPI (Python) – High-performance REST API framework responsible for authentication, business logic, video processing, AI orchestration, and content routing.
- Pydantic – Data validation and request/response modeling for FastAPI.
- Uvicorn – ASGI web server used to host the FastAPI application.

Authentication

- Supabase Authentication – User registration, login, email verification, password reset, and JWT token generation.

Database

- Supabase PostgreSQL – Relational database storing users, videos, moderation results, and application metadata.

Storage

- Supabase Storage – Cloud storage for uploaded workout videos, thumbnails, and media assets.

Artificial Intelligence

- YOLO (You Only Look Once) – Computer vision model that detects people, gym equipment, workout activities, and other visual objects from uploaded videos.
- Pretrained Moderation Model – Evaluates potentially inappropriate or unsafe content and generates moderation scores for the MVP instead of using a custom-trained model.

Business Logic

- Python Rules Engine – Applies configurable moderation policies and determines whether content is published, restricted, sent for manual review, or blocked based on AI results and age policies.

Security

- JWT Authentication – Secures communication between the mobile application and backend APIs using bearer tokens.
- HTTPS – Encrypts all communication between the client, backend, and cloud services.

Deployment

- Expo EAS Build – Builds signed production applications for iOS and Android.
- TestFlight – Distributes beta iOS builds for testing before App Store release.

Scalable Production Architecture

- Load Balancer – Distributes incoming API traffic across multiple FastAPI instances.
- CDN (e.g., Amazon CloudFront) – Delivers videos and static assets through edge locations to reduce latency.
- Containerization (Docker) – Packages backend services for consistent deployment.
- Kubernetes – Orchestrates containers and enables automatic scaling as application demand increases.

Description:

GymTok is a cross-platform social fitness application built with React Native and Expo, enabling a single TypeScript codebase for iOS and Android. The backend is powered by FastAPI, which exposes REST APIs responsible for authentication, business logic, video processing, AI moderation, and content routing. Supabase provides authentication, a PostgreSQL database, cloud storage, and real-time capabilities. When a user uploads a workout video, the backend extracts frames and sends them through a pretrained YOLO computer vision model and moderation pipeline to detect people, gym equipment, workout activities, and potentially unsafe or inappropriate content. A configurable rules engine then assigns moderation scores and automatically routes the content to age-appropriate feeds, an adult-only feed, a manual review queue, or blocks the upload. The architecture is modular and cloud-native, allowing each service—the frontend, backend, AI pipeline, database, and storage—to scale independently from an MVP into a production-scale social media platform.

Architecture

Frontend

React Native + Expo
(User Interface)

|

Backend

FastAPI
(APIs & Business Logic)

|

Cloud Services

Supabase
(Auth, Database & Storage)

|

AI Services

YOLO + Moderation Engine
(Content Analysis & Routing)

|

Personalized Feed

Age-Filtered GymTok Feed

App Screenshots (local test environment for demo purposes, need to integrate supabase for video upload)

